Measurement-Based Quantum Computing (MBQC)
(Session 2 Supplement – Day 1, 3 July 2026)

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Learning Objectives

- Contrast the Measurement-Based Quantum Computing (MBQC) paradigm (One-Way Quantum Computing) with the traditional Circuit Model.
- Understand the structure of resource states, specifically graph and cluster states, and how they are prepared.
- Learn how single-qubit measurements in the XY-plane implement arbitrary rotations and the Hadamard gate.
- Grasp the role of measurement randomness, Pauli by-product (correction) operators, and active feed-forward mechanisms.
- Understand how multi-qubit gates (like CNOT) are realized via measurements on 2D cluster state geometries.

1 Introduction to One-Way Quantum Computing

In the traditional **Circuit Model**, quantum information is stored in qubits that remain static, while computations are performed by applying a sequence of physical, time-dependent unitary gates.

In contrast, **Measurement-Based Quantum Computing (MBQC)**, introduced by Raussendorf and Briegel in 2001, follows a completely different approach:

1. We initialize the computation by preparing a highly entangled, multi-qubit state called a **cluster state** (or graph state). This state is independent of the specific algorithm to be run.
2. The computation is carried out entirely by performing sequential, single-qubit measurements.
3. Qubits are measured in specific bases, and the classical outcomes are used to determine the measurement bases of subsequent qubits.
4. During this process, the entanglement in the resource state is consumed (hence the name **One-Way Quantum Computing**), leaving a set of measured classical bits and a small set of unmeasured output qubits containing the final quantum state.

2 Resource States: Graph and Cluster States

The resource states in MBQC are **graph states**. Let $G = (V, E)$ be a mathematical graph where V represents the set of vertices (qubits) and E represents the set of edges (entangling links).

2.1 Preparation of a Graph State

To prepare the graph state $|G\rangle$ corresponding to G:

1. Initialize all qubits $v \in V$ in the state $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$:

$$|\psi_{\text{init}}\rangle = \bigotimes_{v \in V} |+\rangle_v$$

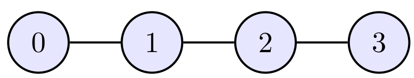
2. For each edge $(u, v) \in E$, apply a Controlled-Z (CZ) gate between qubits u and v:

$$|G\rangle = \left(\prod_{(u,v) \in E} \text{CZ}_{uv} \right) \bigotimes_{v \in V} |+\rangle_v$$

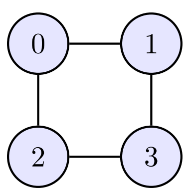
Since CZ gates commute, the order in which they are applied does not matter.

2.2 Standard Geometries

* **1D Cluster State (Qubit Chain):** Used for propagating states and applying single-qubit operations.



* **2D Cluster State (Grid Graph):** Required for universal quantum computation, allowing both single-qubit operations and entangling multi-qubit gates.



3 State Teleportation and Gate Implementation

The fundamental mechanism for propagating information in MBQC is measurement-based state teleportation.

3.1 Single-Qubit State Propagation

Suppose we want to propagate a state $|\psi_{\text{in}}\rangle = a|0\rangle + b|1\rangle$ from qubit 0 to qubit 1.

1. We prepare the 2-qubit entangled state $\text{CZ}(|\psi_{\text{in}}\rangle|+\rangle)$:

$$|\psi_{01}\rangle = \text{CZ}\left((a|0\rangle + b|1\rangle)\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)\right) = \frac{1}{\sqrt{2}}(a|00\rangle + a|01\rangle + b|10\rangle - b|11\rangle)$$

2. We measure qubit 0 in the X-basis $\{|+\rangle, |-\rangle\}$. The projection states are:

$$|\pm\rangle_0 = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}} \implies |0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \quad |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}$$

3. Rewriting the 2-qubit state in the measurement basis of qubit 0:

$$|\psi_{01}\rangle = \frac{1}{2}\left[|+\rangle_0 (a|0\rangle + a|1\rangle + b|0\rangle - b|1\rangle) + |-\rangle_0 (a|0\rangle + a|1\rangle - b|0\rangle + b|1\rangle)\right]$$

$$|\psi_{01}\rangle = \frac{1}{2}\left[|+\rangle_0 ((a+b)|0\rangle + (a-b)|1\rangle) + |-\rangle_0 ((a-b)|0\rangle + (a+b)|1\rangle)\right]$$

Notice that:

$$H|\psi_{\text{in}}\rangle = H(a|0\rangle + b|1\rangle) = \frac{a+b}{\sqrt{2}}|0\rangle + \frac{a-b}{\sqrt{2}}|1\rangle$$

$$XH|\psi_{\text{in}}\rangle = X\left(\frac{a+b}{\sqrt{2}}|0\rangle + \frac{a-b}{\sqrt{2}}|1\rangle\right) = \frac{a-b}{\sqrt{2}}|0\rangle + \frac{a+b}{\sqrt{2}}|1\rangle$$

4. Thus, if the measurement outcome on qubit 0 is $s_0 \in \{0,1\}$ (where $s_0 = 0$ corresponds to $|+\rangle$ and $s_0 = 1$ corresponds to $|-\rangle$), the collapsed state of qubit 1 is:

$$|\psi_{\text{out}}\rangle = X^{s_0} H |\psi_{\text{in}}\rangle$$

The state has been teleported to qubit 1 with an applied Hadamard gate H, up to a **Pauli by-product operator** X^{s_0}.

3.2 Implementing Arbitrary Rotations

To perform a rotation by angle θ, we measure qubit 0 in the rotated basis $\{|\theta_+\rangle, |\theta_-\rangle\}$ in the XY-plane of the Bloch sphere:

$$|\theta_\pm\rangle = \frac{|0\rangle \pm e^{i\theta}|1\rangle}{\sqrt{2}}$$

Measuring in this basis is equivalent to applying an $R_z(-\theta)$ rotation, followed by a Hadamard rotation H and Z-measurement. If the outcome is $s_0 \in \{0,1\}$, the state on qubit 1 becomes:

$$|\psi_{\text{out}}\rangle = X^{s_0} H R_z(-\theta) |\psi_{\text{in}}\rangle$$

4 Feed-forward and Pauli By-product Operators

Because measurement outcomes are probabilistic, the by-product operators (X or Z) accumulate as information propagates down a cluster chain. These random signs must be corrected.

4.1 Active Feed-forward

Consider a 3-qubit chain to apply two rotations: a Z-rotation by θ followed by an X-rotation by ϕ.

1. We measure qubit 0 in the basis $\{|\theta_\pm\rangle\}$, yielding outcome $s_0 \in \{0, 1\}$. The state on qubit 1 is now:
$$|\psi_1\rangle = X^{s_0} H R_z(-\theta) |\psi_{\text{in}}\rangle$$

2. We want to apply an X-rotation $R_x(-\phi) = HR_z(-\phi)H$ on the state. However, the state of qubit 1 has the by-product X^{s_0} applied to it.

3. Because $XR_z(-\phi)X = R_z(\phi)$, the by-product operator changes the direction of the subsequent rotation:
$$R_z(-\phi)X^{s_0} = X^{s_0} R_z\left((-1)^{s_0}(-\phi)\right)$$

4. To compensate for this, we must dynamically adjust the measurement angle of qubit 1 based on Alice's previous outcome s_0:
$$\theta_1 = (-1)^{s_0}\phi$$

5. This mechanism of adjusting future measurement bases based on past measurement outcomes is called **feed-forward**. It is the key to executing deterministic computations in MBQC.

5 Universality in MBQC

To achieve universal quantum computing, we must be able to perform: 1. Arbitrary single-qubit rotations (which we do via 1D cluster chains of length 4). 2. Entangling 2-qubit gates (like CNOT).

The CNOT gate is implemented in MBQC by using a 2D cluster state where two separate 1D qubit chains are bridged by a CZ gate, and sequential measurements are performed to entangle the target and control tracks.

6 Discussion Problems / Exercises

1. **Identity propagation:** Show that propagating a state through a 3-qubit cluster chain $|+\rangle \to |+\rangle \to |+\rangle$ with Z-basis measurements (equivalent to X-basis measurements on the cluster) implements the identity gate I up to Pauli by-products.

   > **Session Takeaway:** Each step along a cluster chain applies a Hadamard rotation: $H \to H \to H \dots$ Propagating through two links applies $H \cdot H = I$, returning the state back to its original basis.

2. **Pauli commutation validation:** Prove that $XR_z(\theta)X = R_z(-\theta)$.

   > **Session Takeaway:** Pauli X flips the direction of Z-rotations. This algebraic relation is the mathematical basis for the feed-forward measurement angle corrections in MBQC.

3. **Euler rotation chain:** How many physical qubits are needed in a 1D cluster state chain to implement an arbitrary single-qubit unitary $U = R_z(\alpha)R_x(\beta)R_z(\gamma)$?

> **Session Takeaway:** We need a chain of 4 qubits (1 input, 3 measured). The three measurements at angles $\theta_0, \theta_1, \theta_2$ implement the three Euler rotations.

7 Take-away Messages

1. MBQC swaps the dynamic control of gates in time for static entanglement in space combined with simple single-qubit measurements.
2. The randomness of quantum measurement is bypassed using classical feed-forward, correcting by-product operators on-the-fly.
3. MBQC is computationally equivalent to the circuit model, making it a highly attractive paradigm for physical platforms where executing gates is hard but preparing large entangled states is natural (e.g. photonics).

Further reading (optional)

Nielsen & Chuang, Section 4.6;
Raussendorf & Briegel, "A One-Way Quantum Computer", PRL 2001;
PennyLane documentation, "Measurement-based quantum computation".

A Detailed Solutions to Discussion Problems

A.1 Solution to Problem 1: Identity Propagation on a 3-qubit Chain

Let the input state on qubit 0 be $|\psi_{\text{in}}\rangle$. We entangle it in a 3-qubit chain (qubits 0, 1, 2). The overall state before measurement is:

$$|\psi_{012}\rangle = \text{CZ}_{12}\text{CZ}_{01}\left(|\psi_{\text{in}}\rangle_0 |+\rangle_1 |+\rangle_2\right)$$

We measure qubits 0 and 1 in the X-basis (equivalent to measuring in the Z-basis after applying Hadamard gates). Let the measurement outcomes be $s_0, s_1 \in \{0, 1\}$. 1. Measuring qubit 0 propagates the state to qubit 1, applying H and the by-product X^{s_0}:

$$\left|\psi_1'\right\rangle = X^{s_0} H |\psi_{\text{in}}\rangle$$

2. Measuring qubit 1 propagates the state to qubit 2, applying another H and the by-product X^{s_1}:

$$\left|\psi_2'\right\rangle = X^{s_1} H \left|\psi_1'\right\rangle = X^{s_1} H \left(X^{s_0} H |\psi_{\text{in}}\rangle\right)$$

Using the relation $HX = ZH$ (since $HXH = Z$), we can commute the H gate past the X^{s_0} operator:

$$\left|\psi_2'\right\rangle = X^{s_1} Z^{s_0} H^2 |\psi_{\text{in}}\rangle$$

Since $H^2 = I$, the final state of the output qubit 2 is:

$$\left|\psi_2'\right\rangle = X^{s_1} Z^{s_0} |\psi_{\text{in}}\rangle$$

The state has been successfully propagated to qubit 2 with zero net logical rotation, up to the Pauli by-product operators $X^{s_1} Z^{s_0}$.

A.2 Solution to Problem 2: Pauli-X and Z-rotation Commutation Relation

We wish to prove:

$$X R_z(\theta) X = R_z(-\theta)$$

Using the definition of the Z-rotation operator:

$$R_z(\theta) = e^{-i\theta Z/2} = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) Z$$

Now we conjugate this operator by X:

$$X R_z(\theta) X = X \left[\cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) Z\right] X$$

Using the linearity of matrix multiplication:

$$X R_z(\theta) X = \cos\left(\frac{\theta}{2}\right) XIX - i \sin\left(\frac{\theta}{2}\right) XZX$$

Since $X^2 = I$, we have $XIX = I$. For the second term, we use the anti-commutation relation of the Pauli matrices ($XZ = -ZX$):

$$XZX = -ZXX = -ZI = -Z$$

Substituting these back:

$$XR_z(\theta)X = \cos\left(\frac{\theta}{2}\right) I + i \sin\left(\frac{\theta}{2}\right) Z$$

Comparing this to $R_z(-\theta)$:

$$R_z(-\theta) = \cos\left(-\frac{\theta}{2}\right) I - i \sin\left(-\frac{\theta}{2}\right) Z = \cos\left(\frac{\theta}{2}\right) I + i \sin\left(\frac{\theta}{2}\right) Z$$

The expressions match exactly, proving the relation.

A.3 Solution to Problem 3: Qubit Count for Euler Rotations

To implement an arbitrary unitary $U = R_z(\alpha)R_x(\beta)R_z(\gamma)$ on a 1D cluster state: 1. Each measurement along a cluster chain applies a rotation and a Hadamard gate:

$$\begin{aligned} &\text{Measurement 0 (angle } \theta_0\text{):} \qquad HR_z(-\theta_0) \\ &\text{Measurement 1 (angle } \theta_1\text{):} \qquad HR_z(-\theta_1) \\ &\text{Measurement 2 (angle } \theta_2\text{):} \qquad HR_z(-\theta_2) \end{aligned}$$

2. The overall operation across three measurements is:

$$U_{\text{net}} = (HR_z(-\theta_2))\,(HR_z(-\theta_1))\,(HR_z(-\theta_0))$$

3. Since $HR_z(\theta)H = R_x(\theta)$, we can rewrite the expression as:

$$U_{\text{net}} = HR_z(-\theta_2)R_x(-\theta_1)R_z(-\theta_0)$$

4. Choosing $\theta_0 = -\gamma$, $\theta_1 = -\beta$, and $\theta_2 = -\alpha$ implements:

$$U_{\text{net}} = HR_z(\alpha)R_x(\beta)R_z(\gamma) = HU$$

5. This implements the desired unitary U followed by a final Hadamard gate H (which can be corrected or accounted for). 6. To perform 3 measurements, we need a chain of 4 qubits: qubit 0 (input), qubits 1 and 2 (measured to propagate and rotate), and qubit 3 (the output qubit). Thus, a 4-qubit chain is sufficient.